MIST 5635-7635: Machine Learning and Business Analytics

Spring 2026

CRN: 61881, 61883, 73239 & 73240

Instructor Information
Name: John Rios
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Email: John.Rios1@uga.edu (please include "ML 1:15 PM" or "ML 2:55 PM" in the subject)
Office Hours (online):

- Tuesday and Thursday, 2:00 - 3:00 pm.
- By appointment, in case the regular office hours do not work for you. Please suggest potential dates/times for the meeting.

Find Zoom meeting info on eLC.

Course Delivery
This course meets in person on Monday and Wednesday at:

- CRN 61881 & 61883: 1:15 pm - 2:35 pm in room 124, Correll Hall
- CRN 73239 & 73240: 2:55 pm - 4:15 pm in room 124, Correll Hall

Course Description and Objectives
Machine learning encompasses a variety of statistical techniques for identifying patterns in data. Companies are increasingly leveraging machine learning to build intelligent systems, anticipate consumer behavior, improve supply-chain efficiency, and more. Given the size and complexity of the data collected by organizations, machine learning is a critical need for many businesses. This course introduces the core concepts and techniques of machine learning, with an emphasis on building predictive models to support data-driven decisions.

The course covers key topics in supervised learning, unsupervised learning, dimension reduction, and feature selection. A variety of methods, such as regression, tree-based models, and deep learning models, will be covered, as well as their implementation in the Python computing environment. A particular emphasis is placed on rigorously training and testing models to achieve high reliability and accuracy. The course will detail the concepts of cross-validation, testing-training splits, and the development of sound machine learning workflows. Finally, the course covers basic principles of deploying and maintaining machine learning pipelines.

Learning Outcomes
Upon successful completion of this course, you will be able to:

Leverage the following concepts in your analysis

- Fundamentals of machine learning, including cross-validation, loss functions, model fit, and feature selection
- Tradeoffs in machine learning design (bias-variance tradeoff, overfitting vs. underfitting, parameter tuning)

- Supervised learning for both regression and classification problems
- Unsupervised learning
- Use of machine learning models for prescriptive applications in business

Apply the following techniques:

- Supervised learning in Python (linear regression, logistic regression, decision trees, ensemble methods, neural networks)
- Unsupervised learning in Python (clustering and dimensionality reduction)
- Conducting cross-validation as part of a machine learning training pipeline

Changes to the Syllabus/Course
The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.

Updates, changes, or clarifications to the syllabus, assignments, or other course-related matters will typically be addressed at the beginning or end of class. I also occasionally post updates to the eLC website and send notifications via eLC's email function or to your official UGA email address. You are responsible for updates via any of these mechanisms.

Prerequisites, co-requisites
MIST 4600, 4600E, or equivalent with a minimum grade of C.

Material
Software: We will be using the Python programming environment for all coursework and assignments. It is expected that students in the course will have a basic knowledge of the Python language, and students are required to have the most recent version downloaded (https://www.python.org/downloads/) on their computer. All course notes and assignments will be given as Jupyter Notebooks (https://jupyter.org/install). There are multiple ways to run a Jupyter Notebook; a straightforward option is to install Anaconda (https://www.anaconda.com/download) and launch the notebook locally. Other resources will be made available on the course eLC page.

Textbook: The main text for this course will be *Introduction to Machine Learning with Python* by Andreas C. Müller and Sarah Guido (link). A pdf copy can be found online and will also be posted on eLC. The accompanying GitHub page for the textbook can be found here: link. That site will be useful for downloading the code/datasets directly.

Recommended Texts:
- A solid (but technical) supplemental text for this course will be *Introduction to Statistical Learning with Applications in Python*, by James et al. The website for this text is here: (https://www.statlearning.com/). From the website, you can download a free copy of the second edition. This book covers much of the same topics as our text, but with less focus on applied ML.
- For an excellent perspective on applying ML concepts in business settings, I recommend *Data Science for Business* by Foster Provost and Tom Fawcett (link). Several lectures will

actually draw on materials from this book, but the text itself is not required for any particular assignment.

File Security

You are responsible for the security of your files. To avoid problems, I recommend regularly making backup copies of your work. In addition, when modeling or visualizing data in R, save your files to prevent data loss in the case of a computer crash.

Active Learning Statement

This is an exciting field that is continually evolving, and we are all here to learn, including the instructor. We will learn from the course material, your practice, classmates, guest speakers, and the news, and we will achieve this goal through a flipped classroom and Active Learning's dynamic interactions and conscious reflections.

UGA Student Honor Code

"*I will be academically honest in all of my academic work and will not tolerate academic dishonesty of others.*" A Culture of Honesty, the University's policy and procedures for handling cases of suspected dishonesty, can be found at honesty.uga.edu.

UGA Well-being Resources:

UGA Well-being Resources promote student success by cultivating a culture that supports a more active, healthy, and engaged student community.

Anyone needing assistance is encouraged to contact Student Care & Outreach (SCO) in the Division of Student Affairs at 706-542-8479 or visit sco.uga.edu. Student Care & Outreach helps students navigate difficult circumstances by connecting them with the most appropriate resources or services. They also administer the Embark@UGA program which supports students experiencing, or who have experienced, homelessness, foster care, or housing insecurity.

UGA provides both clinical and non-clinical options to support student well-being and mental health, any time, any place. Whether on campus, or studying from home or abroad, UGA Well-being Resources are here to help.

- Well-being Resources: well-being.uga.edu
- Student Care and Outreach: sco.uga.edu
- University Health Center: healthcenter.uga.edu
- Counseling and Psychiatric Services: caps.uga.edu or CAPS 24/7 crisis support at 706-542-2273
- Health Promotion/ Fontaine Center: healthpromotion.uga.edu
- Accessibility & Testing: https://accessibility.uga.edu/

Additional information, including free digital well-being resources, can be accessed through the UGA app or by visiting https://well-being.uga.edu.

Inclusive Excellence
The Terry College of Business is committed to promoting an inclusive learning and working environment among its students, faculty, and staff. This class welcomes the open exchange of ideas and values freedom of thought and expression and provides a professional environment that recognizes the inherent worth of every person. It aims to foster dignity, understanding, and mutual respect among all individuals in the class.

Professional Development (MIS Undergraduate Only)
The course has the following professional development requirements to help you prepare for the job market and long-term professional success:

- Two events from the following: MIS Department events (e.g., MIS Welcome Back Social) or MIS Department student organization events (e.g., SMIS, WiT). You must register for the event a minimum of 24 hours in advance.

OR

- Take an MIS-related online course that furthers your knowledge. For example, you might take a Python course online, and successfully complete the course exam, possibly receiving a certificate. Get approval from your instructor before you take the course. A course that is shown to take one hour counts for one event, two hours for two events, and so on.

- Attend events or complete activities that are consistent with the course that you are taking. For example, if you are taking MIST 5620, attend a Society of Business Intelligence (SBI) presentation.

- There is no "double counting" of attendance at events. Each event attended can only count for a single course. Within MISTERS, select for which course you want your attendance to count.

- If you have potential conflicts with meeting the professional development requirements or think there are better development activities for your situation, meet with your instructor to discuss the possibilities. This meeting must be at the start, not the end of the semester and is your responsibility to schedule. Requests for alternative ways of meeting the professional development requirements must be entered into MISTERS by the end of 13th week.

- You may want to include the following on your resume: "I have completed XXX hours of professional development, including attending multiple professional meetings, networking with industry professionals, and completing LinkedIn Learning courses."

Grading and Deliverables

Item	Points	Graduate/honor points
Quizzes	16	16
Problem sets	33	33
TechTalks	10	10
Midterm I	20	20
Midterm II	20	20
Course evaluation	1	1
Professional development	2	-
Total**	**102***	**100**

* Will be adjusted to a score out of 100

** Your final points score for the class is based on the weights in the preceding table. Do not use eLC's computed score.

Quizzes. There will be *11 quizzes* during the semester based on in-class exercises and material. Approximately once a week, a short quiz will be posted on eLC related to the outcomes of any in-class exercises, coding fundamentals, or class concepts. The quizzes should be straightforward to complete for students who attend class and participate in activities. Quizzes must be completed by the beginning of the next class period. Students can *submit answers twice* before the deadline, with only incorrect answers repeated on the second attempt. The quiz score will be the *average of two attempts*. The *lowest quiz score from the semester will be dropped*.

Problem Sets. There will be *six problem sets* assigned during the semester that require students to use Python to apply machine learning concepts learned in class. Each problem set will be accompanied by a data set provided by the instructor or as part of the required libraries. Successful completion of the assignment involves uploading the data, conducting analyses, and reporting the results. Refer to the syllabus and eLC for exact due dates. All students are expected to turn in their own assignment; if students receive help in any form (e.g., other students, the internet, LLMs), it *MUST be documented* in the submission.

Exams. There will be two midterm exams during the semester. These will be a mix of multiple-choice questions, free-response questions, and (potentially) brief applications in Python.

Graduate Course Credit. For students enrolled in 7635, there are some additional elements that must be completed to receive the graduate course credit, including extra questions on problem sets and exams.

Participation. Students are expected to participate in class and come prepared for the hands-on lab sessions. Attendance is expected (but not formally required) unless cleared with the instructor beforehand. Notice that Quizzes are an important component of your course grade, and they are based on the in-class activities.

If you cannot complete an exercise on time or take an exam at the specified time, please advise the instructor as soon as possible so that alternative arrangements can be made.

There is an **optional cumulative final exam** for the course. If you intend to take the optional final exam, you must inform the instructor of this via email no later than **5:00 pm, April 24th, 2026**. *If you take the cumulative final exam, then the midterm exams will count for 13% each, and the final exam will count for 14% of your final grade*.

Use of AI for Coursework
Unless explicitly stated, artificial intelligence-based technologies, such as ChatGPT, must not be used to generate responses for student assignments. Unauthorized use of artificial intelligence software or word-mixing software to write your paper or disguise plagiarized work is considered unauthorized assistance in this course.

Grade Ranges
Course grades are assigned using the following ranges:

- A = 93 or above
- A- = 90 to 92.99
- B+ = 87 to 89.99
- B = 83 to 86.99
- B- = 80 to 82.99
- C+ = 77 to 79.99
- C = 73 to 76.99
- C- = 70 to 72.99
- D = 60 to 69.99
- F = 59.99 or below

Grades will not be rounded up. For example, an 89.99 is a B+ and will not be changed to a 90.

Schedule

Date	Topic	Assignment due date @ 11:59pm the day before	Reference (IMLP)
Mon 1-12	Course Intro and Syllabus	-	
Wed 1-14	Basics – The What and Why of ML		Ch. 1 (all)
Mon 1-19	**Martin Luther King Jr. Day- No Classes**		
Wed 1-21	Basics – The ML Process		
Mon 1-26	Basics – Over & Under Fitting		Ch. 2.2
Wed 1-28	Feature Engineering and Data Description		Ch. 4-4.4
Mon 2-2	Feature Engineering and Data Description		Ch. 4.5-4.6
Wed 2-4	Supervised Learning – Intro and Datasets		Ch. 2.3-2.3.1, Ch. 5.3
Mon 2-9	Supervised Learning – Linear Regression	PS 1	Ch. 2.3.3
Wed 2-11	Supervised Learning – Regularized Regression		Ch. 2.3.3
Mon 2-16	Supervised Learning – Logistic Regression		Ch. 2.3.3
Wed 2-18	Supervised Learning – Decision Trees		Ch. 2.3.5
Mon 2-23	Supervised Learning – Ensembles	PS 2	Ch. 2.3.6
Wed 2-25	Midterm Review		Ch. 2.3.6
Mon 3-2	**Midterm 1**	PS3	
Wed 3-4	Supervised Learning – Neural Networks		Ch. 2.3.8
Mon 3-9	**Spring Break - No Classes**		
Wed 3-11	**Spring Break - No Classes**		
Mon 3-16	Supervised Learning – Wrap Up		Ch. 2.5
Wed 3-18	ML Pipelines – Cross Validation (CV)		Ch. 5-5.1
Mon 3-23	ML Pipelines – CV & Grid Search		Ch. 5.2
Wed 3-25	ML Pipelines – Developing Pipelines		Ch. 6 (all)
Mon 3-30	Unsupervised Learning – Intro and Dataset		Ch. 3-3.3
Wed 4-1	Dimensionality Reduction	PS 4	Ch. 3.4
Mon 4-6	Unsupervised Learning – K-Means		Ch. 3.5.1
Wed 4-8	Unsupervised Learning – Clustering Variants		Ch. 3.5.2-3.5.5
Mon 4-13	Text Analysis – Intro and Handling Text Midterm Review	PS 5	Ch. 7-7.3
Wed 4-15	**Midterm 2**		
Mon 4-20	Text Analysis – Bag of Words and Topic Modeling		Ch. 7.4-7.9
Wed 4-22	Text Analysis – Advanced Text Prediction		
Mon 4-27	ML Deployment, Monitoring, and Responsible Use Midterm Review	PS6	
	Final Exam: Scheduled by The Office of the Registrar		